Checkpoint A - full solutions

Full solutions

Use these solutions only after completing the 60-minute attempt. Review each method, result and common mistake, then record the first cause of any lost mark.

Question 1 — reasoning, algebra and one-variable calculus

Part A — `CPA-Q01-PA` (`M1`, `M2`) Method and conditions. Quantify both variables over the stated domains. To negate a universally quantified equality, change both universal quantifiers to existential quantifiers and negate the equality. A counterexample must satisfy $n \geq 1$ and $r \neq 1$.

The claim is

$$\forall n \in \{1, 2, ...\} \; \forall r \in \mathbb{R} \;\; \{1\} : \quad \sum_{j=0}^{n} r^j = \frac{1 - r^n}{1 - r}.$$

Its logical negation is

$$\exists n \in \{1, 2, ...\} \; \exists r \in \mathbb{R} \;\; \{1\} : \quad \sum_{j=0}^{n} r^j \neq \frac{1 - r^n}{1 - r}.$$

For example, take $n = 1$ and $r = 2$. The left-hand side is $1 + 2 = 3$, whereas the claimed right-hand side is

$$\frac{1 - 2}{1 - 2} = 1.$$

This admissible counterexample disproves the claim. The correct identity is

$$\sum_{j=0}^{n} r^j = \frac{1 - r^{n+1}}{1 - r}, \qquad r \neq 1.$$

Final result. The claim is false; its exponent should be $n + 1$, not n.

Marks (4). Universal statement: 1. Logical negation: 1. Admissible counterexample with both sides evaluated: 1. Correct finite-sum identity: 1.

Common mistakes. Negating the equality without changing the quantifiers; using the excluded value $r = 1$; keeping the incorrect exponent n in the corrected identity.

Part B — `CPA-Q01-PB` (`M3`) Method and conditions. The polynomial is continuous on the closed interval $[-2, 4]$, so a global maximum exists. Compare its values at every interior stationary point and both endpoints.

Differentiate and factor:

$$f'(x) = -3x^2 + 6x + 9 = -3(x - 3)(x + 1).$$

The stationary points in the interval are therefore $x = -1$ and $x = 3$. The candidate values are

x	-2	-1	3	4
$f(x)$	2	-5	27	$20.$

Final result. The global maximum is

$$\max_{x\in[-2,4]} f(x) = 27,$$

attained at $x = 3$.

Marks (5). Correct derivative and factorisation: 1. Both stationary points: 1. Values at the two stationary points: 1. Values at both endpoints: 1. Correct global maximum and location: 1.

Common mistakes. Checking only stationary points; omitting an endpoint; declaring a global maximum without comparing all candidates.

Question 2 — linear systems and multivariable differentiation

Part A — `CPA-Q02-PA` (`M4`) Method and conditions. Apply each row operation to the entire augmented row. Compare the coefficient rank with the number of unknowns after elimination.

Write the augmented matrix and eliminate:

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 3 \\ 2 & -1 & 1 & 4 \\ 0 & 1 & 1 & 2 \end{array}\right] \xrightarrow{R_2 \leftarrow R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 1 & 0 & 3 \\ 0 & -3 & 1 & -2 \\ 0 & 1 & 1 & 2 \end{array}\right].$$

Next use $R_2 \leftarrow R_2 + 3R_3$:

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 3 \\ 0 & 0 & 4 & 4 \\ 0 & 1 & 1 & 2 \end{array}\right].$$

Thus $z = 1$, then $y = 1$, and then $x = 2$. There are three pivots in the coefficient columns, so $\operatorname{rank}(\mathbf{A}) = 3$. The augmented matrix also has rank 3.

Final result. The system has the unique solution

$$(x, y, z) = (2, 1, 1).$$

Marks (4). Traceable valid elimination/row-operation evidence showing three pivots: 1. Correct solution: 1. Correct coefficient rank: 1. Unique-solution classification justified by rank and the number of unknowns: 1. Substitution-only work does not earn the first mark.

Common mistakes. Applying a row operation only to the coefficient entries; omitting the rank statement; classifying the system from the number of equations rather than its pivots.

Part B — `CPA-Q02-PB` (`M6`) Method and conditions. The maps are polynomial or linear, so they are differentiable everywhere. Under the column-gradient convention,

$$\nabla(F \circ \mathbf{G})(\mathbf{x}) = \mathbf{J}_{\mathbf{G}}(\mathbf{x})^{\mathsf{T}} \nabla F(\mathbf{G}(\mathbf{x})).$$

Here

$$\nabla F(u, v) = \begin{bmatrix} 2u + v \\ u \end{bmatrix} \in \mathbb{R}^{2\times 1}, \qquad \mathbf{J}_{\mathbf{G}}(x, y) = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \in \mathbb{R}^{2\times 2}.$$

Since

$$\mathbf{G}(1,0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad \nabla F(1,1) = \begin{bmatrix} 3 \\ 1 \end{bmatrix},$$

the chain rule gives a 2×1 column vector:

$$\nabla h(1,0) = \mathbf{J}_{\mathbf{G}}(1,0)^{\mathsf{T}} \nabla F(1,1) = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}.$$

For the requested direct check, first simplify the composition:

$$h(x,y) = (x+y)^2 + (x+y)(x-y) = 2x^2 + 2xy.$$

Therefore

$$\frac{\partial h}{\partial x} = 4x + 2y, \qquad \frac{\partial h}{\partial x}(1,0) = 4,$$

which agrees with the first component above.

Final result. $\nabla h(1,0) = [4,2]^{\mathsf{T}}$.

Marks (4). Correct derivative objects, evaluation point and dimensions: 1. Dimensionally correct chain-rule setup: 1. Correct column gradient: 1. Correct direct verification of the first component: 1.

Common mistakes. Writing the gradient as a row; reversing the chain-rule multiplication order; evaluating ∇F at $(1,0)$ instead of at $\mathbf{G}(1,0)$; omitting the direct check.

Question 3 — probability rules and Bayes' rule

Part A — `CPA-Q03-PA` (`S1`) Method and conditions. Use inclusion–exclusion. The exactly-one event excludes the intersection, and the neither event is $(A \cup B)^c$.

First,

$$\mathbb{P}(A \cup B) = \frac{3}{5} + \frac{1}{2} - \frac{1}{4} = \frac{17}{20}.$$

The probability of exactly one event is

$$\mathbb{P}(A \cup B) - \mathbb{P}(A \cap B) = \frac{17}{20} - \frac{1}{4} = \frac{3}{5}.$$

Finally,

$$\mathbb{P}\big((A \cup B)^c\big) = 1 - \frac{17}{20} = \frac{3}{20}.$$

Final result. The three probabilities are 17/20, 3/5 and 3/20, respectively.

Marks (3). One mark for each correct probability with a valid setup.

Common mistakes. Adding the event probabilities without subtracting their overlap; including the intersection in exactly one; using the complement of $A \cap B$ for neither.

Part B — `CPA-Q03-PB` (S2) Method and conditions. The events H and H^c partition the sample space. Use the law of total probability for S, then apply Bayes' rule. Here $\mathbb{P}(H^c) = 3/4$.

The two positive-screen branches are

$$\mathbb{P}(S \cap H) = \mathbb{P}(S \mid H)\mathbb{P}(H) = \frac{4}{5}\frac{1}{4} = \frac{1}{5},$$

and

$$\mathbb{P}(S \cap H^c) = \mathbb{P}(S \mid H^c)\mathbb{P}(H^c) = \frac{1}{5}\frac{3}{4} = \frac{3}{20}.$$

Therefore

$$\mathbb{P}(S) = \frac{1}{5} + \frac{3}{20} = \frac{7}{20}.$$

Bayes' rule gives

$$\mathbb{P}(H \mid S) = \frac{\mathbb{P}(S \mid H)\mathbb{P}(H)}{\mathbb{P}(S)} = \frac{1/5}{7/20} = \frac{4}{7}.$$

Final result. $\mathbb{P}(S) = 7/20$ and $\mathbb{P}(H \mid S) = 4/7$.

Marks (4). Correct two branch probabilities: 1. Correct total $\mathbb{P}(S)$: 1. Correct Bayes setup and direction: 1. Correct posterior: 1.

Common mistakes. Reversing the conditional probability; omitting the H^c branch; using $\mathbb{P}(H)$ rather than $\mathbb{P}(S)$ in the denominator.

Question 4 — random variables and moments

Part A — `CPA-Q04-PA` (S3) Method and conditions. A PMF must sum to one over its stated support. The CDF includes its endpoint: $F_X(1) = \mathbb{P}(X \leq 1)$.

Normalisation gives

$$c(1 + 2 + 3) = 1,$$

so $c = 1/6$. Hence

$$\mathbb{P}(X \geq 1) = p_X(1) + p_X(2) = \frac{2}{6} + \frac{3}{6} = \frac{5}{6},$$

and

$$F_X(1) = p_X(0) + p_X(1) = \frac{1}{6} + \frac{2}{6} = \frac{1}{2}.$$

Final result. $c = 1/6$, $\mathbb{P}(X \geq 1) = 5/6$, and $F_X(1) = 1/2$.

Marks (3). Correct normalising constant: 1. Correct upper-tail probability: 1. Correct CDF value with the endpoint included: 1.

Common mistakes. Failing to normalise the PMF; starting the stated support at one; using $\mathbb{P}(X < 1)$ for $F_X(1)$.

Part B — `CPA-Q04-PB (S4)` Method and conditions. Use linearity of expectation and the variance formula for a linear combination. The given finite variances and covariance are sufficient.

The expectation is

$$\mathbb{E}[Z] = 2\mathbb{E}[X] - \mathbb{E}[Y] = 2(1) - 2 = 0.$$

The variance is

$$\begin{aligned} \operatorname{Var}(Z) &= 4\operatorname{Var}(X) + \operatorname{Var}(Y) - 4\operatorname{Cov}(X, Y) \\ &= 4(2) + 3 - 4\left(\frac{1}{2}\right) \\ &= 9. \end{aligned}$$

Incorrectly treating X and Y as independent would omit the covariance term and give $8 + 3 = 11$. The positive covariance lowers the variance by 2 because the coefficients of X and Y in Z have opposite signs.

Final result and interpretation. $\mathbb{E}[Z] = 0$ and $\operatorname{Var}(Z) = 9$; the positive covariance lowers this variance from the incorrect independence value of 11.

Marks (3). Correct expectation: 1. Correct variance setup and value: 1. Correct comparison and sign interpretation: 1.

Common mistakes. Ignoring the covariance term; failing to square the coefficient (2); using a positive covariance contribution despite the opposite-signed coefficients.